CS 170 - HW 1 - Jie Wen

1 Study Group

none

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2 Course Policies

(a) The exams for CS170 this semester are scheduled as follows:

- **Midterm 1:** Tuesday, 9/30/2025, 7:00 PM - 9:00 PM.
- **Midterm 2:** Tuesday, 11/4/2025, 7:00 PM - 9:00 PM.
- **Final:** Tuesday, 12/16/2025, 8:00 AM - 11:00 AM.
  There are no planned alternate or remote exams.

(b) Although homework is due on Saturdays at 10:00 PM with a late deadline at 11:59 PM, it is recommended that I have my homework finished by 10:00 PM. This allows the late deadline to be a buffer for technical issues.

(c) If I miss the Gradescope late deadline, my homework will not be accepted, even in cases of technical issues or emergencies. The course provides two automatic homework drops to account for such situations.

(d) The primary source of communication for CS170 is Edstem. All important announcements and course updates will be posted there, and students are responsible for staying up-to-date.

(e) I have read and understood the course syllabus and policies.

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3 Understanding Academic Integrity

Based on the course policies, here is the evaluation of each statement:

(a) **Not OK**.

(b) **Not OK**.

(c) **Not OK**.

(d) **OK**.

(e) **Not OK**.

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4 Math Review

(a) Simplify into a single log

(i) $\dfrac{\ln x}{\ln y} = \log_y x$.

(ii) $\ln x + \ln y = \ln(xy)$.

(iii) $\ln x - \ln y = \ln\left(\dfrac{x}{y}\right)$.

(iv) $170 \ln x = \ln\left(x^{170}\right)$.

(b) Proofs

(i) Show $x^{\log_{1/x} y} = \dfrac{1}{y}$.

Let

$$z = \log_{1/x} y$$

By definition of logarithm:

$$\left(\frac{1}{x}\right)^z = y \quad \Rightarrow \quad x^{-z} = y \quad \Rightarrow \quad x^z = \frac{1}{y}$$

(ii) Show $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Pairing argument: write the sum forwards and backwards

$$S = 1 + 2 + \cdots + n, \qquad S = n + (n-1) + \cdots + 1.$$

Adding termwise gives $2S = (n+1) + (n+1) + \cdots + (n+1) = n(n+1)$. Thus $S = \frac{n(n+1)}{2}$.

(iii) Sum of a geometric progression: $\sum_{k=0}^{n} ar^k$.

Let $S = a + ar + \cdots + ar^n$. If $r \neq 1$, multiply by r:

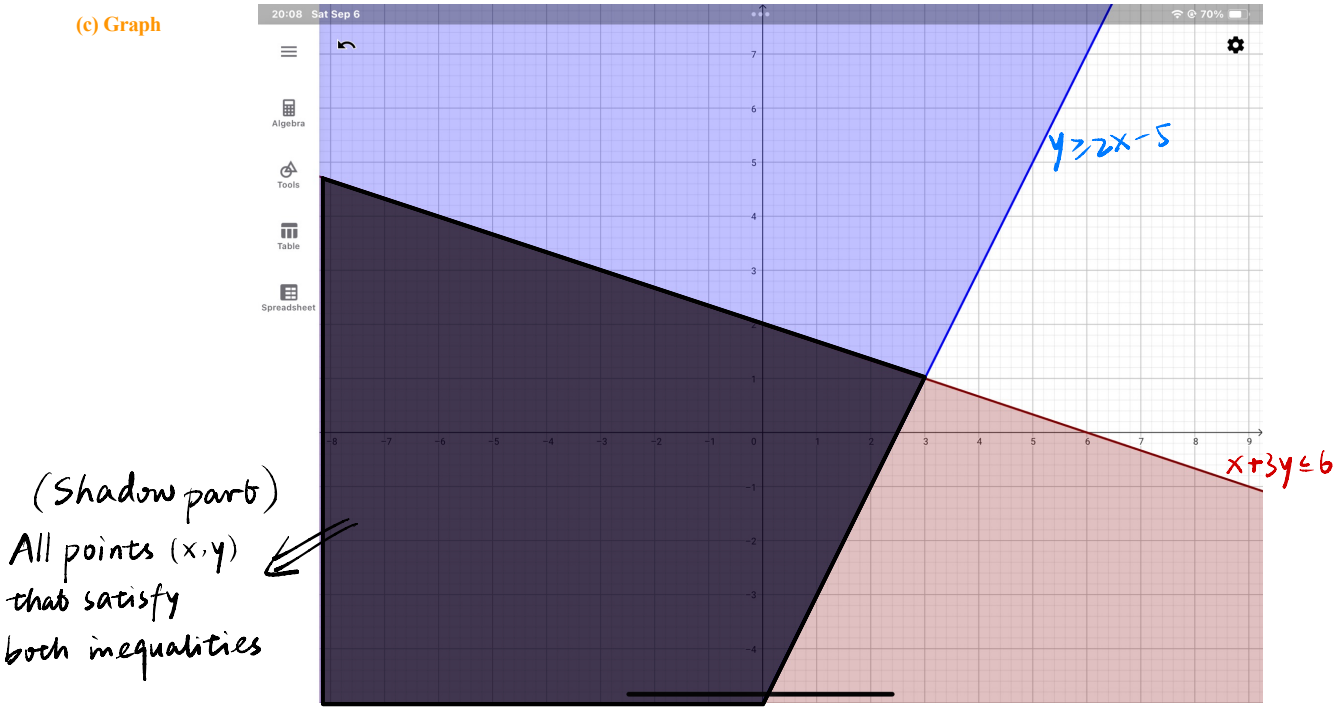
$$rS = ar + ar^2 + \cdots + ar^{n+1}.$$

Subtract: $S - rS = a - ar^{n+1}$, so $(1-r)S = a(1-r^{n+1})$. Therefore

$$S = \frac{a(1-r^{n+1})}{1-r}, \qquad (r \neq 1).$$

If $r = 1$, every term equals a and there are $n+1$ terms, so $S = a(n+1)$.

(c) Graph



5 Asymptotics Practice

(a) Answers

f	g	$f = O(g)$	$g = O(f)$	**Both**
$n^2 + 5n$	$1000(n+1)^2$	✓	✓	✓
$5n^3$	$n^3 + (\log n)^{10}$	✓	✓	✓
n^{100}	1.01^n	✓		
$(\log n)^{100}$	$n^{0.1}$	✓		
$n \cdot 2^n$	3^n	✓		
$n!$	n^n	✓		

(b) Justified answer for $f(n) = 1 + b + \cdots + b^n$ and $g(n) = b^n$, $b > 0$.

Case 1: $0 < b < 1$.
The geometric series converges to a constant:

$$f(n) = \frac{1 - b^{n+1}}{1 - b} \to \frac{1}{1 - b} \quad \text{as } n \to \infty,$$

while $g(n) = b^n \to 0$. Thus $g(n) = O(f(n))$ (indeed $g(n) \to 0$ so $g(n) = o(f(n))$), but $f(n)$ is not $O(g(n))$ because $f(n)$ stays bounded away from 0 while $g(n)$ tends to 0.

Case 2: $b = 1$**.**
Then $f(n) = 1 + 1 + \cdots + 1 = n + 1$ and $g(n) = 1$. So $g(n) = O(f(n))$ (a constant is $O(n)$), but $f(n)$ is not $O(g(n))$ (linear growth cannot be bounded by a constant).

Case 3: $b > 1$**.**
Sum formula gives

$$f(n) = \frac{b^{n+1} - 1}{b - 1} = \frac{b}{b - 1} b^n - \frac{1}{b - 1}.$$

Hence for large n,

$$\frac{b}{2(b - 1)} b^n \le f(n) \le \frac{2b}{b - 1} b^n$$

(for example, choose n large enough that b^{n+1} dominates the -1). Therefore $f(n) = \Theta(b^n)$, so $f = O(g)$ and $g = O(f)$ (**both**).

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6 Recurrence Relations

(a) $T(n) = 2T(n/3) + 5n$

Analyze the recurrence by unrolling it:

- **Level 0**: cost = $5n$
- **Level 1**: cost = $2 \cdot 5(n/3) = \frac{10}{3}n$
- **Level 2**: cost = $4 \cdot 5(n/9) = \frac{20}{9}n$
- **Level k**: cost = $2^k \cdot 5(n/3^k) = 5n\left(\frac{2}{3}\right)^k$

The recursion stops when $n/3^k \le 1$, i.e., $k \approx \log_3 n$. So the total cost is:

$$T(n) = 5n \sum_{k=0}^{\log_3 n} \left(\frac{2}{3}\right)^k$$

This is a geometric series:

$$\sum_{k=0}^{\log_3 n} \left(\frac{2}{3}\right)^k = \frac{1 - (2/3)^{\log_3 n + 1}}{1 - 2/3} = 3\left[1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}\right]$$

Thus,

$$T(n) = 15n\left[1 - \left(\frac{2}{3}\right)^{\log_3 n + 1}\right] = 15n - 10n \cdot n^{\log_3(2) - 1}$$

Since $\log_3(2) < 1$, the term $n^{\log_3(2)}$ grows slower than n. Therefore, the dominant term is $15n$, and conclude:

$$\boxed{T(n) = \Theta(n)}$$

(b) Algorithm A

The algorithm:

- Partitions the input into 5 subproblems of size $n/5$
- Recursively solves 3 of them
- Partitioning time is $O(n \log n)$

This gives the recurrence:

$$T(n) = 3T(n/5) + O(n \log n)$$

Unrolling the recurrence:

- **Level 0**: cost = $O(n \log n)$
- **Level 1**: cost = $3 \cdot O\left(\frac{n}{5} \log \frac{n}{5}\right) = O(n \log n)$
- **Level k**: cost = $3^k \cdot O\left(\frac{n}{5^k} \log \frac{n}{5^k}\right) = O\left(n\left(\frac{3}{5}\right)^k \log n\right)$

The number of levels is $\log_5 n$. So the total cost is:

$$T(n) = O(n) \sum_{k=0}^{\log_5 n} \left(\frac{3}{5}\right)^k (\log n - k \log 5)$$

As $n \to \infty$, the sum converges to a constant multiple of $\log n$. Therefore:

$$\boxed{T(n) = \Theta(n \log n)}$$

(c) $T(n) = T(3n/5) + T(4n/5), T(1) = 1$**.**

First find a candidate form. Try $T(n) = \Theta(n^b)$. Plugging $T(n) \approx cn^b$ into the recurrence gives

$$cn^b \approx c((\frac{3}{5})^b + (\frac{4}{5})^b)n^b.$$

So now we want $(\frac{3}{5})^b + (\frac{4}{5})^b = 1$. Test small integers:

- $b = 1$: $(\frac{3}{5}) + (\frac{4}{5}) = 1.4 > 1$.
- $b = 2$: $(\frac{3}{5})^2 + (\frac{4}{5})^2 = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1$.

So $b = 2$ exactly satisfies the balance equation. That suggests $T(n) = \Theta(n^2)$.

Rigorous sandwich (induction) argument.

Because the recurrence is homogeneous, if for all smaller m we have $am^2 \leq T(m) \leq bm^2$ for constants $0 < a \leq b$, then for n

$$T(n) = T(3n/5) + T(4n/5) \leq b\left(\frac{3n}{5}\right)^2 + b\left(\frac{4n}{5}\right)^2 = bn^2\left(\tfrac{9}{25} + \tfrac{16}{25}\right) = bn^2.$$

And similarly

$$T(n) \geq a\left(\frac{3n}{5}\right)^2 + a\left(\frac{4n}{5}\right)^2 = an^2.$$

So if the inequalities hold up to smaller sizes, they hold for n as well. Choose a and b to accommodate the finite base cases (for example, $a = \min_{1 \leq m \leq m_0} T(m)/m^2$ and $b = \max_{1 \leq m \leq m_0} T(m)/m^2$ for some small m_0). That closes the induction, giving constants $a, b > 0$ so that for all n,

$$an^2 \leq T(n) \leq bn^2.$$

Thus

$$\boxed{T(n) = \Theta(n^2).}$$

(d) $T(n) = T(\sqrt{n}) + 1$**, with** $T(1) = T(2) = 1$

Let $n = 2^{2^m}$. Then $\sqrt{n} = 2^{2^{m-1}}$. Define:

$$S(m) = T(2^{2^m})$$

Then:

$$S(m) = S(m-1) + 1, \quad S(0) = T(2) = 1$$

This gives:

$$S(m) = m + 1$$

Since $m = \log_2(\log_2 n)$:

$$\boxed{T(n) = 1 + \log_2(\log_2 n) = \Theta(\log \log n)}$$